

Information loss and minimal feedback in teacher–student policy learning: a Hamilton–Jacobi analysis

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Abstract

Constrained teacher actions can discard future-cost information needed by a shared student policy. Hamilton–Jacobi policy evaluation identifies the missing normal component; the student’s output space determines which part matters. For a known isotropic quadratic model and a fixed output space, we characterize the minimum number of exact scalar value-difference queries preserving all student cost comparisons. Feasible queries attain this rank bound. Direct student-basis queries and a simple hybrid are also exact: fewer queries than the hybrid require rank below both the normal and student dimensions. Across 480 neural fits and 120 direct-interface checks, recovered information reproduces full-information learning. Random embeddings give no saving over the hybrid; a constructed shared-actuation geometry requires one query instead of three. A spectral bound quantifies approximate alignment and response error. Separately, a 600-fit loss intervention confirms lower Bellman regret despite higher action MSE under severe constraints, with heterogeneous effects, adverse conditions and uncertain closed-loop benefit. Exact failure examples and nonlinear transfer studies delimit what preserved teaching information can guarantee.

Keywords: Teacher–student learning, Hamilton–Jacobi–Bellman equation, Policy improvement, Sensitivity learning, Constrained control

1. Introduction

A teacher can supply an improving action at every training state while a student fitted to those actions becomes worse. Input constraints can hide the cost of deviating from a target, and a shared policy must trade off deviations across states. We ask which teacher information must be retained for that tradeoff to preserve cost comparisons.

The setting is deterministic feedback control with known differentiable dynamics and a common cost. Hamilton–Jacobi (HJ) policy evaluation defines the teacher’s future-cost sensitivity; Hamilton–Jacobi–Bellman (HJB) improvement uses it to compare actions. The teacher need not be optimal. Converting its improvement signal into supervision introduces two obstructions. Constrained targets can be identical despite opposite cost effects of their best shared regression fit. Nonempty improvement sets can also be jointly unrealizable, even when the student class contains a policy that improves average cost. We give exact examples with complete coverage and globally solved fitting.

For a known isotropic quadratic teaching model, the missing information is a normal component at the constrained target. A student restricted to a fixed output space needs only the projection of that component into its own space. We prove that the minimum number of exact scalar value-difference queries is $\text{rank}(P^\top \Gamma)$, where P spans student outputs and Γ spans normal ambiguity, and construct feasible queries attaining it. A gain scalar suffices on a ball but can fail on a box. Direct student-basis queries also recover exactly. A hybrid using the smaller of normal and student dimensions already attains the bound for generic relative position; strict savings require additional alignment. A spectral truncation bound describes the approximation error when that alignment is perturbed.

Cost-aware imitation, Hamiltonian fitting and improvement from imperfect oracles are established in AggreVaTeD, MPC-Net and MAMBA [1, 2, 3]. Related foundations include guided policy search, distillation and regularized improvement [4, 5, 6, 7], constrained HJB policy iteration [8], viscosity-based operator learning [9] and proximal policy distillation [10]. Constraint-induced ambiguity is studied in inverse reinforcement learning [11]; the kernel criterion used here belongs to classical linear information recovery [12, 13]. Our contribution is the specific information requirement obtained by combining teacher constraint geometry with student representation.

The evidence separates exact information recovery from policy improvement. A 480-fit grid and 120 direct-interface checks compare feasible queries with full information, student-space controls and a simple hybrid. A con-

structured shared-actuation case tests positive rank below both ambient dimensions; random embeddings identify where there is no advantage over the hybrid. A separate 600-fit loss intervention tests normal restoration, scalar weighting, shuffled information and constraint/capacity controls. Its initial primary is inconclusive; independent confirmation finds lower Bellman regret despite higher action MSE, with heterogeneous effects and uncertain closed-loop benefit. Conditional transfer bounds, joint-realizability examples and the full nonlinear audit and set studies are supplied in the appendices.

2. Hamilton–Jacobi formulation

For deterministic dynamics $\dot{x} = f(t, x, u)$, running cost ℓ and terminal cost g , a fixed feasible teacher ϕ has value

$$V^\phi(t, x) = \int_t^T \ell(s, x_s^\phi, \phi(s, x_s^\phi)) ds + g(x_T^\phi), \quad J(\phi) = \mathbb{E}_{x_0} V^\phi(0, x_0). \quad (1)$$

In a smooth region it satisfies policy evaluation,

$$V_t^\phi + H(t, x, \phi, \nabla V^\phi) = 0, \quad H(t, x, u, p) = \ell(t, x, u) + p^\top f(t, x, u). \quad (2)$$

The optimal HJB equation instead minimizes over actions:

$$\partial_t V^* + \min_{u \in U} H(t, x, u, \nabla_x V^*) = 0, \quad V^*(T, x) = g(x). \quad (3)$$

Proposition 1. *Suppose V^ϕ is continuously differentiable along the relevant trajectories, both feasible policies have well-defined trajectories and integrable costs, and they share the initial distribution and terminal cost. Then*

$$J(\pi) - J(\phi) = \mathbb{E} \int_0^T \mathcal{A}^\phi(t, x_t^\pi, \pi(t, x_t^\pi)) dt, \quad (4)$$

$$\mathcal{A}^\phi(t, x, u) = H(t, x, u, \nabla V^\phi) - H(t, x, \phi(t, x), \nabla V^\phi). \quad (5)$$

Proof. The chain rule and teacher evaluation equation give $\ell(t, x_t^\pi, \pi) + \frac{d}{dt} V^\phi(t, x_t^\pi) = \mathcal{A}^\phi(t, x_t^\pi, \pi)$. Integrate and use $V^\phi(T, \cdot) = g$. $\square$

Nonpositive integrated advantage implies no greater student cost. The condition concerns student-visited states, which teacher samples alone need not cover.

For $x_{t+1} = F_t(x_t, u_t)$, the discrete counterpart is

$$A_t^\phi(x, u) = c_t(x, u) + V_{t+1}^\phi(F_t(x, u)) - V_t^\phi(x), \quad (6)$$

$$J(\pi) - J(\phi) = \mathbb{E}_{x_0} \sum_{t=0}^{H-1} A_t^\phi(x_t^\pi, \pi_t(x_t^\pi)). \quad (7)$$

Differentiating the fixed teacher's feedback trajectory gives

$$p_t = c_x + (D_x \phi_t)^\top c_u + (F_x + F_u D_x \phi_t)^\top p_{t+1}, \quad p_H = \nabla g. \quad (8)$$

One reverse sweep obtains the covectors without a state-space grid or fitted scalar critic [14, 15]. At projection boundaries, automatic differentiation uses the implemented branch.

2.1. Feasible targets

For control-affine dynamics and isotropic quadratic action cost, write

$$F_t(x, u) = a_t(x) + B_t u, \quad c_t(x, u) = c_{0t}(x) + r_t \|u\|^2 + l_t^\top u, \quad b_t = l_t + B_t^\top p_{t+1}. \quad (9)$$

The local Bellman model is $q_t(u) = r_t \|u\|^2 + b_t^\top u$, up to an action-independent constant. For Euler dynamics $F = x + hf$ and $c = h\ell$, q_t/h is the control-dependent Hamiltonian with frozen next-state covector. For closed convex U and $r_t > 0$,

$$u_t^+ = \arg \min_{u \in U} \{q_t(u) + \frac{\rho}{2} \|u - u_t^\phi\|^2\}, \quad (10)$$

$$= \text{proj}_U \frac{\rho u_t^\phi - b_t}{2r_t + \rho}. \quad (11)$$

Anisotropic costs require metric projection. The same local expansion is used at the final decision; its terminal remainder need not vanish. For fixed state and teacher action, nonexpansivity gives

$$\|u^+(\hat{p}) - u^+(p)\| \leq \frac{\|B^\top(\hat{p} - p)\|}{2r + \rho}. \quad (12)$$

2.2. Information lost at constraints

Put $Q_\rho = q + \rho \|u - u^\phi\|^2/2$ and $a = r + \rho/2$. Then

$$Q_\rho(u) - Q_\rho(u^+) = a \|u - u^+\|^2 + \nabla Q_\rho(u^+)^\top (u - u^+). \quad (13)$$

The normal term is nonnegative on U and vanishes for an interior target. For a ball, if both actions lie on its sphere and $\nabla Q_\rho(u^+) = -\lambda u^+$, the difference becomes $(a + \lambda/2) \|u - u^+\|^2$. General feasible students require the additional term derived below.

2.3. Identical labels, different costs

Proposition 2. *Exact one-step Bellman models can have identical strictly improving targets and identical globally optimal target fits with opposite teacher-relative cost effects, even with common quadratic curvature, complete coverage and a teacher in the student class.*

Proof. Take three equiprobable contexts, $U = [-1, 1]$, teacher zero and constant students $u = c$. Let

$$Q_i(u) = (u - \mu_i)^2, \quad (\mu_1, \mu_2, \mu_3) = (\tau, -1, -1), \quad \tau \geq 1. \quad (14)$$

For $\rho = 0$, the targets are $(1, -1, -1)$ and each improves its teacher. Their unique minimum-MSE fit is $c = -1/3$, with MSE $8/9 < 1$. However,

$$J(-1/3) - J(0) = \frac{2\tau - 3}{9}, \quad (15)$$

which is negative at $\tau = 1$ and positive for $\tau > 3/2$.

An exact Bellman realization uses $x = (s, y)$, $F(s, y, u) = (s, u)$, $c_t = u^2$, and $g(s, y) = \mu(s)^2 - 2\mu(s)y$, with $\mu(s) = -1 + (\tau + 1)(s - 2)(s - 3)/2$. Start uniformly at $(s, 0)$, $s \in \{1, 2, 3\}$. Total cost is Eq. (14); the projected terminal covector is $-2\mu_i$. Sensitivity error and continuation remainder both vanish. $\square$

At $\tau = 10$, the discarded normal term is $18(1 - u)$ in the first context and zero elsewhere. Teacher cost is 34, best target-fit cost is $323/9 = 35.888889$, and direct cost minimization in the same constant class selects $c = 1$ with cost $89/3 = 29.666667$. For any fixed finite $\rho \geq 0$, replacing μ_i by $\beta(\tau, -1, -1)_i$, $\beta = 1 + \rho/2$, keeps the proximal targets unchanged and gives fitted cost difference $[1 + 2\beta(\tau - 2)]/9 > 0$ at $\tau = 10$. This is an existence construction for each fixed ρ .

Retaining $a_z = r_z + \rho_z/2$ and $n_z = \nabla Q_{\rho,z}(u_+)$ restores

$$L_{\text{cost}}(u; z) = a_z \|u - u_+\|^2 + n_z^\top (u - u_+) = Q_{\rho,z}(u) - Q_{\rho,z}(u_+). \quad (16)$$

For a fixed state distribution and a class containing the teacher, a global minimizer cannot increase mean proximal model value. The teacher's zero proximal penalty gives the same conclusion for the unregularized model. This is a policy-cost comparison in the exact one-step example; nonlinear continuation and changed occupancy require the bounds in Appendix A.

On $U = \{u : \|u\| \leq R\}$, write $n_z = -\lambda_z u_+$, $\lambda_z \geq 0$, with $\lambda_z = 0$ for interior targets. Expansion gives

$$L_{\text{cost}}(u; z) = (a_z + \lambda_z/2) \|u - u_+\|^2 + \frac{\lambda_z}{2} (R^2 - \|u\|^2). \quad (17)$$

Weighted action fitting omits the second term. It can therefore miss the cost of a student’s interior compromise even with the correct weight.

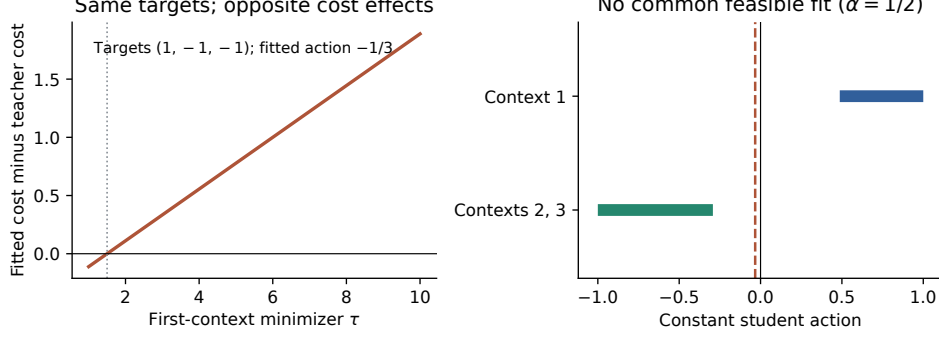


Figure 1: Exact constructed failures of the teaching interface. Left: the same constrained action targets and the same best constant fit have opposite cost effects as the unconstrained minimizer τ changes, with quadratic curvature fixed. Right: each improvement set is nonempty, but their intersection is empty; the dashed line is the best constant set-distance fit (Appendix E.4).

3. Minimal teaching information

The student need not recover the entire normal vector in Eq. (16). We characterize the part affecting its cost comparisons under a specified scalar-feedback interface.

3.1. Interface

At a fixed teacher state, let

$$Q(u) = a\|u\|^2 + b^\top u + c, \quad v = \arg \min_{u \in U} Q(u), \quad n = 2av + b \in -N_U(v). \quad (18)$$

Here $a > 0$, U is a centered ball or box of radius or half-width $R > 0$, and $N_U(v) = \{w : w^\top(u - v) \leq 0 \text{ for all } u \in U\}$. The receiver knows a, U, v and orthonormal $P \in \mathbb{R}^{d \times m}$; its actions lie in $S \cap U$, $S = \text{range}(P)$. It has no access to b, n . A feasible query returns $Q(u) - Q(0)$, giving

$$y(u) = Q(u) - Q(0) - a\|u\|^2 + 2av^\top u = n^\top u. \quad (19)$$

We count additional scalar responses, with target and baseline supplied. The bound excludes arbitrary encoded messages, derivative access and learners allowed to recompute b from the full analytic model. This is an abstract

restricted teaching interface, not a demonstrated communication bottleneck in a deployed controller.

Let orthonormal $\Gamma \in \mathbb{R}^{d \times k}$ span $K = \text{span}(-N_U(v))$; $k = 0$ for interior targets. For student actions, only $\beta = P^\top n$ matters:

$$Q(Pz) - Q(Pz') = a(\|z\|^2 - \|z'\|^2) + (-2aP^\top v + \beta)^\top (z - z'). \quad (20)$$

Normal differences in $K \cap S^\perp$ are therefore irrelevant. The recovery requirement covers every feasible pair in $S \cap U$; it is stronger than recovering only the comparisons used by a particular fitted network.

Proposition 3. *Fixed feasible queries with row matrix W determine every cost difference in $S \cap U$, uniformly over models sharing a, U, v , iff*

$$\ker(W\Gamma) \subseteq \ker(P^\top \Gamma). \quad (21)$$

The minimum worst-case count is

$$r = \text{rank}(P^\top \Gamma) \leq \min(k, m). \quad (22)$$

This lower bound includes deterministic adaptive queries and is attained by a feasible nonadaptive construction.

Proof. Write $n = \Gamma\eta$. Observations $W\Gamma\eta$ determine $P^\top \Gamma\eta$ precisely under Eq. (21). Since $S \cap U$ contains a neighborhood of zero in S , a coefficient difference changes a feasible cost difference. Necessity holds within the normal cone: small opposite perturbations of a relative-interior normal along an unobserved direction in K remain admissible and, by strong convexity, preserve the unique target v .

With $q < r$ queries, the row space of $W\Gamma$ cannot contain that of $P^\top \Gamma$. For adaptive queries, follow the transcript at a relative-interior normal and perturb within the common kernel of those queries. Both perturbations preserve every response and subsequent query, but change the student coefficient.

For attainment, choose orthonormal $E \in \mathbb{R}^{m \times r}$ spanning $\text{range}(P^\top \Gamma)$. Query $u_j = \alpha P E_j$, $\alpha = R/2$, where R is the ball radius or box half-width. Each query has norm α and is feasible. The responses satisfy $y = \alpha E^\top \beta$, so $\beta = E y / \alpha$. If $r = 0$, no query is required. $\square$

3.2. Direct-query controls

For any orthogonal $H \in \mathbb{R}^{m \times m}$, the m feasible queries $u_j = \alpha P H_j$ give $y = \alpha H^\top \beta$ and $\beta = H y / \alpha$. Thus both the student coordinate basis and a random orthogonal student basis recover exactly. A hybrid choosing

full-normal recovery when $k < m$ and student-basis recovery otherwise uses $\min(k, m)$ responses. It already attains the bound whenever $r = \min(k, m)$; relevant queries then offer no count advantage. Interior targets need no query for any method.

A strict query-count advantage over the hybrid occurs when $r < \min(k, m)$. We focus on the nontrivial case $r > 0$. For example, let $d = 8$, $K = \text{span}(e_1, e_2, e_3)$ and

$$P = [g, e_4, e_5, e_6]H, \quad g = (e_1 + e_2 + e_3)/\sqrt{3}, \quad (23)$$

with a generic orthogonal 4×4 matrix H . Here $k = 3, m = 4, r = 1$. All three active coordinate normals and all four student coordinate directions are visible, so deleting zero directions alone does not attain the bound. The student can act on only their shared combination. Recognizing this dependency, by any equivalent rank-revealing method, reduces three hybrid queries to one.

3.3. Learning rule

At each cached state, form Γ, E from U, v, P , collect the r responses, and train $u = Pz_\theta(x)$ with

$$L_{\text{relevant}}(z; x) = a\|z\|^2 + (-2aP^\top v + Ey/\alpha)^\top z = Q(Pz) - Q(0). \quad (24)$$

This preserves the full-information objective and its parameter gradients where differentiable, without reconstructing n or requiring the student to realize v . Response error e gives

$$\|\widehat{\beta} - \beta\| \leq \frac{\|e\|}{\alpha}, \quad |\widehat{\Delta Q} - \Delta Q| \leq \frac{\|e\|}{\alpha} \|z - z'\|. \quad (25)$$

The result is statewise and requires known curvature, exact geometry and fixed P ; ordinary hidden width does not imply a fixed output space. Equation (25) is a local error bound.

3.4. Approximate alignment

Exact rank can increase under arbitrarily small perturbations of P . If a bound $\|n\| \leq M$ is available, a truncated construction instead gives a continuous error tradeoff. Let E_q contain the first q left singular vectors of $P^\top \Gamma$, padding its singular values σ_j with zeros beyond the spectrum. Querying $\alpha P E_q$ and decoding noisy responses as above yields

$$\|\widehat{\beta} - \beta\|^2 \leq M^2 \sigma_{q+1}^2 + \|e\|^2 / \alpha^2. \quad (26)$$

Indeed, $\beta = P^\top \Gamma \eta$, $\|\eta\| = \|n\| \leq M$; the omitted singular components have norm at most $M\sigma_{q+1}$ and are orthogonal to the decoded response error. Multiplying the square root of this bound by $\|z - z'\|$ bounds the cost-difference error. This is an approximation upper bound requiring M , not an exact recovery or approximate-query optimality claim. Numerical rank tolerances must therefore be interpreted against the discarded spectrum.

3.5. Ball and box constraints

At an active ball target, $n = -\lambda v$, $k = 1$. If $P^\top v \neq 0$, one scalar is necessary and sufficient. The target gain gives

$$G = Q(0) - Q(v) = (a + \lambda)R^2, \quad \lambda = G/R^2 - a. \quad (27)$$

It recovers the full radial-slack loss, whereas scalar weighting of MSE omits that slack. If $P^\top v = 0$, no additional information is needed.

For a box with k active coordinates, Γ contains their coordinate vectors and generically $r = \min(k, m)$. Gain alone can fail: with $a = 1$, $U = [-1, 1]^2$, $b_A = (-3, -5)$ and $b_B = (-5, -3)$ both yield $v = (1, 1)$ and $G = 6$. Yet costs relative to zero at $(1, 0), (0, 1)$ are $(-2, -4)$ for A and $(-4, -2)$ for B .

In the HJ model, b contains $B^\top \nabla V^\phi$; the rank thus counts hidden future-cost directions relevant to this student. For an approximate continuation model, recovery preserves that model; policy transfer still requires the stated remainder and occupancy bounds.

4. Controlled neural experiments

The two neural studies use horizon 12 and the exact system

$$x_{t+1} = 0.8x_t + 0.2u_t, \quad u_t \in U, \quad c_t = 0.36\|x_t\|^2 + 0.05\|u_t\|^2, \quad g = \|x_{12}\|^2. \quad (28)$$

Teacher $\phi = 0$ has $V_t^\phi(x) = \|x\|^2$ and advantage

$$A_t^\phi(x, u) = a\|u\|^2 + b(x)^\top u, \quad a = 0.09, \quad b(x) = 0.32x. \quad (29)$$

The coefficient includes quadratic continuation 0.2^2 . The target is $v = \text{proj}_U(-16x/9)$, with normal $n = 2av + b(x)$.

Except in the specified shared-actuation construction, initial states have probability 0.1 of a randomly signed first coordinate of magnitude uniform on $[6, 10]$ and others uniform on $[-0.1, 0.1]$. Otherwise the first coordinate is uniform on $[-0.1, 0.1]$ and others on $[-1, 1]$. This anisotropic law is common to all radii; changing R also changes the feasible control problem.

Each fit uses 1,024 training initials and their 12 teacher states, 512 independent validation initials and 4,096 test initials. Paired methods share initialization, data and minibatches. Adam uses 4,096 updates, batch 512, learning rate 0.003, moments (0.9, 0.999), epsilon 10^{-8} and norm clipping at 10. Zero output weights initialize the student at the teacher. The final checkpoint is saved before test evaluation, without selection, gate or roll-back. Validation curves are descriptive. Fits run sequentially on an RTX 4080 SUPER.

The local endpoint on held-out teacher states is

$$\mathcal{R}(\pi) = \frac{\mathbb{E}[A^\phi(x, \pi(x)) - A^\phi(x, v(x))]}{\mathbb{E}[-A^\phi(x, v(x))]} \quad (30)$$

Zero realizes the target; one equals teacher regret. Full 12-step costs are evaluated separately on student trajectories. The studies use different output parameterizations and are analyzed separately.

5. Minimal-feedback experiment

We test Proposition 3 on the system in Section 4, with $d \in \{8, 32\}$, balls or boxes, $R \in \{0.5, 2\}$ and $m \in \{2, 4\}$. The student is a bias-free $d \rightarrow 32 \rightarrow m$ SiLU network followed by fixed orthonormal P . Uniform scaling by latent Euclidean norm for balls or embedded maximum absolute coordinate for boxes enforces feasibility while preserving $\text{range}(P)$. Parameters are float32; geometry and coefficients are float64.

5.1. Interfaces and protocol

The 480-fit grid uses five seeds and six interfaces: target only (set the unknown normal to zero), gain (one scalar per active target), relevant (r student-visible queries), random (r ambient normal-space queries with minimum-norm decoding), full face (k normal-basis queries), and privileged full information. Gain uses the minimum-norm normal consistent with the response and known face. All methods fit Eq. (24), normalized by ad .

Four direct controls are evaluated on all 80 caches: student basis, random orthogonal student basis, hybrid, and pruned hybrid. The last deletes normal coordinate directions orthogonal to the student and student coordinate directions orthogonal to the normal space, then chooses the smaller remaining basis. It does not combine linearly dependent directions. Direct controls add 80 fits using the original five paired seeds in four representative conditions (Table 1). These conditions were chosen after inspection of the

grid; they check interface fidelity and are not an independent performance confirmation.

Another 40 fits use five new seeds, eight interfaces (the four direct controls plus target, relevant, full face and full information), and Eq. (23) with $d = 8, m = 4, R = 0.5$. The first three initial coordinates have independent random signs and magnitudes uniform on $[6, 10]$; the others are uniform on $[-0.1, 0.1]$. All 12 teacher states have exactly those three active coordinates. This is a constructed shared-actuation test of $k = 3, m = 4, r = 1$, not a naturally occurring benchmark. A latent rotation makes all student axes visible.

Protocols, seeds and numerical-source hashes are fixed before their respective fits. The primary checks exact coefficient recovery to relative RMS tolerance 10^{-9} and actual scalar query counts. Secondary regret fidelity uses tolerance 10^{-3} per pair against full information, without a population equivalence claim. All 600 fits are retained. The scalar oracle excludes shared target/baseline preparation; full-vector derivatives use a different interface. The analytic coefficient is hidden by experimental restriction only.

5.2. Results

All coefficient and count checks pass. The 480-fit grid reproduces full-information regret within 3.81×10^{-12} in all 80 pairs. All 120 additional checkpoints pass independent re-evaluation; the 110 exact-recovery/full-information comparisons differ in regret by at most 8.04×10^{-13} and in closed-loop cost by 2.83×10^{-11} . The maximum coefficient error across all direct checks is 3.56×10^{-15} . This is the expected consequence of preserving the objective, not evidence of a better optimizer.

Every random-embedding cache has $r = \min(k, m)$. In the box caches, relevant, hybrid and pruned hybrid each use 332,362 responses; student basis and its orthogonal rotation use 444,234; full face uses 1,039,456. Thus the 68.03% reduction against full-face recovery is shared by the simple hybrid. Relevant queries save 25.18% against the direct student basis and *nothing against hybrid*. Ball counts likewise match hybrid. The ambient-random baseline’s failures do not extend to a complete orthogonal student-space basis.

In the shared-actuation construction, all 61,440 cached states have $k = 3, m = 4, r = 1$. Relevant feedback uses one query per state, hybrid and pruned hybrid three, and either student basis four (Figure 2). Every recovery method yields mean regret 0.523674 and mean closed-loop cost 189.698903, versus 0.627840 and 190.883229 for target-only fitting. The extra saving

depends on linear dependence among visible directions, not simply zero-coordinate pruning. The construction establishes this possibility, not its prevalence in applications.

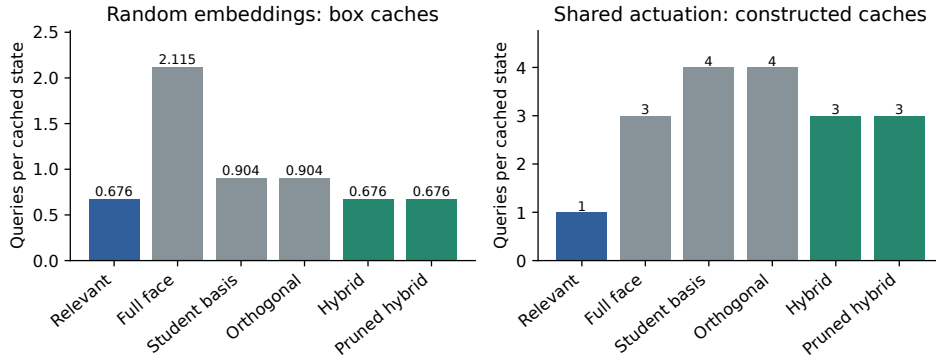


Figure 2: Direct controls separate dimension reduction from additional alignment. Counts include inactive targets. Left: all random-embedding box caches; relevant and hybrid coincide. Right: a constructed shared actuation geometry; every state has positive rank one.

Table 1: Additional interface checks: means over five paired seeds. All exact-recovery interfaces agree to the reported precision; full-reference values are shown. These checks reuse four conditions and add a constructed shared-actuation condition.

Condition	Fits	Regret	Cost	Max. $ \Delta\mathcal{R} $
box, $d = 8, m = 2, R = 0.5$	20	0.585439	8.101119	0
box, $d = 32, m = 2, R = 0.5$	20	0.900619	15.475675	0
box, $d = 32, m = 4, R = 2$	20	0.820332	15.014408	0
ball, $d = 32, m = 4, R = 0.5$	20	0.715371	15.313708	8.03×10^{-13}
Shared actuation	40	0.523674	189.698903	0

Timing includes geometry, query construction, actual responses and decoding on 12,288-state batches, after one warmup and with three shuffled repeats. Median batch times on random box caches are 1.348 ms for relevant, 0.753 ms for student basis and 0.890 ms for hybrid; on structured caches they are 2.268, 0.879 and 0.802 ms. With this cheap analytic oracle, rank construction costs more than the saved responses. These implementation measurements do not support a runtime advantage.

The complete grid, including gain/random failures and uncertain target-only comparisons, is in Appendix B and Appendix C. A further 2,304 algebraic checks verify Eq. (26): arbitrarily small perturbations can raise exact

rank from one to three, while one-query approximation error changes continuously with the omitted singular values (Appendix D). No noisy neural training is claimed.

6. Neural loss intervention

This separate study varies constraints, capacity and loss information in the eight-dimensional system of Section 4, with ball constraints. Its learned output matrix is not the fixed embedding required by Proposition 3.

6.1. Student

The student is a bias-free $8 \rightarrow w \rightarrow 8$ SiLU network with radial projection, $w \in \{2, 4, 16\}$. Actions lie in its output matrix’s column span. Width 16 can represent the target exactly by $\text{SiLU}(s) - \text{SiLU}(-s) = s$, verified numerically. Zero output weights initialize every student at the teacher; first-layer weights are paired across losses.

6.2. Protocol

Up to common division by action dimension, the losses are

$$L_{\text{point}} = \|u - v\|^2, \quad (31)$$

$$L_{\text{restored}} = \|u - v\|^2 + (n/a)^\top (u - v), \quad (32)$$

$$L_{\text{weight}} = (1 + \|n\|/(2aR))\|u - v\|^2, \quad (33)$$

$$L_{\text{shuffled}} = \|u - v\|^2 + (n_\sigma/a)^\top (u - v). \quad (34)$$

The fixed random permutation σ breaks the state-normal association. Restoration equals Bellman regret divided by a ; weighting omits radial slack. At $R = 32$ all normals vanish.

The initial study crosses $R \in \{0.5, 2, 8, 32\}$, three widths, four losses and five seeds (240 fits), with restored-minus-point at $R = 2, w = 2$ as primary. Confirmation selects the initial severe-constraint secondary condition $R = 0.5, w = 2$ as primary, with ten independent seeds, $R \in \{0.5, 2, 32\}$, all widths and losses (360 fits). Its plan, source hashes and seeds are frozen before training. Samples are not pooled. Paired 95% t intervals use seed means and are not multiplicity-adjusted.

6.3. Results

The initial primary is inconclusive: regret difference $+0.000318$ with 95% interval $[-0.016912, 0.017548]$ and one win in five pairs. The severe-constraint secondary improves in all five pairs at widths 2 and 4, though the width-2 interval crosses zero.

Independent confirmation at $R = 0.5, w = 2$ reduces normalized regret from 0.549027 to 0.448915 (18.23%). The paired difference is -0.100112 $[-0.198038, -0.002185]$, with ten wins. Eight effects lie between -0.03884 and -0.03238 ; two are -0.37530 and -0.34355 . The width-4 secondary also has ten wins but interval $[-0.139534, 0.002312]$.

In the confirmation primary, action MSE rises from 0.020590 to 0.022357 while the normal contribution falls from 0.047222 to 0.034574. Exact regret, their sum with MSE multiplied by $8a$, falls from 0.062047 to 0.050671. Weight-only regret is 0.504262; restored minus weight is -0.055347 $[-0.057134, -0.053560]$. Weighting achieves smaller weighted-distance loss (0.036247 versus 0.042683) but larger omitted slack (0.020673 versus 0.007989). Shuffled-normal regret is 0.635812, with restored-minus-shuffled difference -0.186897 $[-0.247484, -0.126310]$. These are post-fit decompositions, without a claim of unique causal mediation.

At $R = 32$, parameter tensors are identical across losses (135 paired checks). At $R = 0.5, w = 16$, confirmation regrets are 0.000298 for point and 0.000309 for restored. At $R = 2, w = 2$ the secondary difference is -0.010305 $[-0.018903, -0.001706]$, whereas $R = 2, w = 4$ is adverse: $+0.017940$ $[0.004030, 0.031849]$. All conditions appear in Table 2.

Confirmation-primary closed-loop cost falls from 8.204291 to 8.103785 (1.23%), with ten wins but interval $[-0.232391, 0.031378]$. Both means beat teacher cost 8.713862, so this experiment does not reproduce the exact witness’s teacher-worsening outcome. Rare-state cost falls from 61.154090 to 60.134279; common-state cost rises from 2.225335 to 2.232217. Restored-minus-weight cost is -0.025450 $[-0.026580, -0.024321]$. All 600 saved policies reproduce reported means; decomposition errors are below 3.1×10^{-14} and telescoping errors below 7.4×10^{-14} .

Figure 4 shows all ten confirmation-primary paired effects. Their mean is -0.10011 and median -0.03639 ; the two largest reductions account for 71.8% of the summed reduction. All signs favor restoration, but the typical paired effect is smaller than the mean. The prespecified mean and its interval remain the primary analysis; this descriptive plot exposes heterogeneity.

Table 2: Controlled neural information transfer. Mean normalized local Bellman regret; lower is better. Δ is restored minus point with an unadjusted paired 95% t interval across seeds. The initial primary is $R = 2$, width 2; the independent confirmation primary is $R = 0.5$, width 2. All radii and widths are shown, including adverse comparisons.

R	Width	Point	Restored	Weight	Shuffled	Δ [95% CI]
<i>Initial study: 5 seeds, 240 runs</i>						
0.5	2	0.6216	0.4545	0.5110	0.6286	-0.1671 [-0.3809, +0.0467]
0.5	4	0.4609	0.2638	0.3208	0.3895	-0.1971 [-0.3748, -0.0194]
0.5	16	0.0003	0.0005	0.0003	0.0560	+0.0002 [+0.0000, +0.0004]
2	2	0.4069	0.4072	0.4104	0.4960	+0.0003 [-0.0169, +0.0175]
2	4	0.2725	0.2885	0.2914	0.2890	+0.0161 [-0.0105, +0.0427]
2	16	0.0002	0.0004	0.0003	0.0041	+0.0002 [-0.0000, +0.0003]
8	2	0.2768	0.2790	0.2772	0.2777	+0.0023 [-0.0036, +0.0081]
8	4	0.1960	0.1939	0.1960	0.1961	-0.0021 [-0.0077, +0.0036]
8	16	0.0004	0.0004	0.0003	0.0006	+0.0000 [-0.0000, +0.0001]
32	2	0.2691	0.2691	0.2691	0.2691	+0.0000 [+0.0000, +0.0000]
32	4	0.1924	0.1924	0.1924	0.1924	+0.0000 [+0.0000, +0.0000]
32	16	0.0004	0.0004	0.0004	0.0004	+0.0000 [+0.0000, +0.0000]
<i>Independent confirmation: 10 seeds, 360 runs</i>						
0.5	2	0.5490	0.4489	0.5043	0.6358	-0.1001 [-0.1980, -0.0022]
0.5	4	0.3302	0.2615	0.3220	0.4057	-0.0686 [-0.1395, +0.0023]
0.5	16	0.0003	0.0003	0.0003	0.0376	+0.0000 [-0.0001, +0.0002]
2	2	0.4038	0.3935	0.3995	0.4170	-0.0103 [-0.0189, -0.0017]
2	4	0.2667	0.2846	0.2852	0.2870	+0.0179 [+0.0040, +0.0318]
2	16	0.0002	0.0003	0.0002	0.0033	+0.0001 [+0.0000, +0.0002]
32	2	0.3252	0.3252	0.3252	0.3252	+0.0000 [+0.0000, +0.0000]
32	4	0.1870	0.1870	0.1870	0.1870	+0.0000 [+0.0000, +0.0000]
32	16	0.0004	0.0004	0.0004	0.0004	+0.0000 [+0.0000, +0.0000]

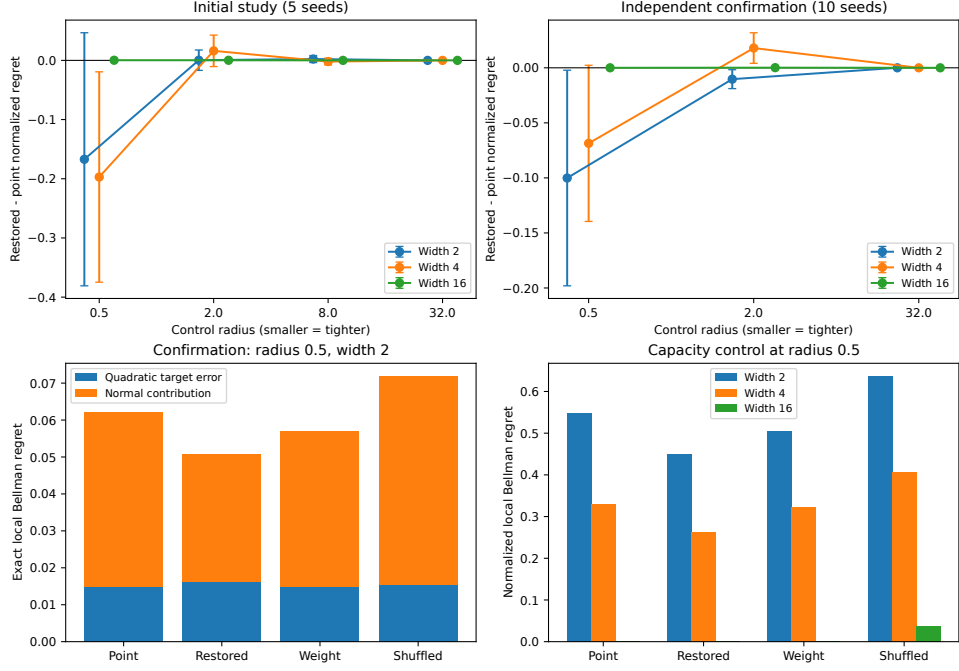


Figure 3: Top: restored-minus-point local regret and paired 95% t intervals for the initial and independent samples. Bottom: confirmation loss decomposition on held-out teacher states and capacity response.

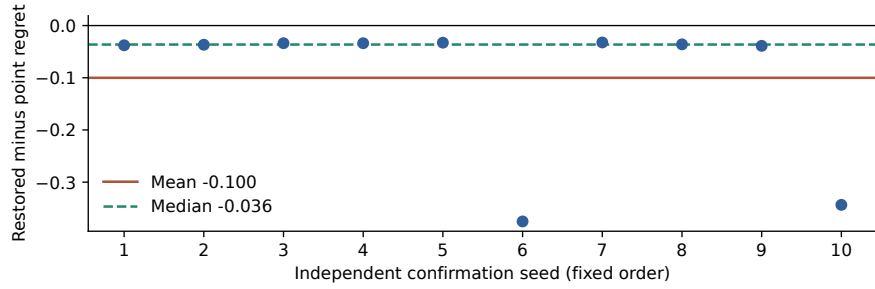


Figure 4: All ten independent confirmation-primary paired differences in normalized Bellman regret. Lines mark the mean and median.

7. Transfer to policy cost

Preserving teacher-state cost comparisons does not certify an updated policy. Summed advantages must be evaluated on student trajectories; approximate continuation additionally requires sensitivity and curvature control. The bounds in Appendix A separate target gain, fitting error, sensitivity discrepancy and continuation remainder. Local improvement sets do not remove the shared-policy constraint: an exact three-context example has nonempty sets with no common student, although average-cost improvement is attainable (Appendix E).

Separate nonlinear studies test these limits. In 50 continuation-audit fits, the guided policy reduces cost relative to its initialization by 7.74% and 18.64% on mechanics and reaction tasks, but remains 0.42% and 2.60% above its teacher. The comparison does not establish an advantage over adaptive regularization. Sampled fitting costs can exceed local target gains, and 26 of 4,800 perturbed-action checks violate the empirical curvature bound (Appendix G–Appendix H).

A 56-fit improvement-set study retains an independent five-seed comparison: mean cost is 0.54% below point-target fitting, with a paired interval crossing zero, and 1.28% above upper-model fitting. All five selected set policies remain worse than their teacher (Appendix I). These different procedures neither establish general optimization superiority nor identify the same causal mechanism as the controlled loss intervention.

8. Discussion

HJ policy evaluation supplies the teacher’s future-cost covector; constraints can hide its action-relevant normal component in an otherwise exact improving target. A shared student then makes cost tradeoffs without the information needed to evaluate them. The exact witness isolates this failure without derivative error, missing states or imperfect fitting. The loss intervention tests its consequence in neural learning, with a conditional and heterogeneous restoration effect.

The query theorem specifies how much of the hidden component matters to a fixed student. Its proof uses classical linear recovery, and its learning objective is established cost-aware fitting. Direct basis queries and hybrid recovery are equally valid implementations. The additional insight is the joint role of constraint and student geometry: generic position needs $\min(k, m)$ responses, whereas dependencies among visible directions can reduce the count

further. The constructed experiment and spectral bound show respectively exact and approximate forms of this relation.

Known isotropic curvature, a fixed embedding and the restricted scalar oracle are substantial assumptions. Given the analytic plant and value gradient, a learner can bypass the oracle. Our experiments do not demonstrate that such restricted access is needed in practice, and they show no runtime saving with cheap values. Exact rank savings can disappear under perturbation; approximate recovery needs a normal-norm bound as well as the discarded spectrum. Learned critics, changing output spaces and noisy neural fitting remain untested.

The neural loss study uses linear dynamics and an anisotropic state law. Its null initial primary, adverse condition, uneven confirmation effects and uncertain closed-loop mean delimit its practical claim. The nonlinear audit and set studies retain empirical bound violations and policies worse than their teachers. Preserved local information does not remove fitting, occupancy or continuation requirements.

9. Conclusion

The HJ view identifies cost distinctions that constrained teacher labels can erase. Under the stated quadratic interface, the student’s projected normal rank is exactly the scalar information needed to preserve all its cost comparisons. A simple hybrid is optimal in generic position; additional query savings require lower-rank alignment, with approximate error controlled by the omitted singular spectrum. Neural experiments verify the information accounting and a conditional restoration effect, while separate transfer studies leave general policy improvement unresolved.

Research assistance

OpenAI Codex assisted with formulation, code, experiments, analysis and writing.

Data and code availability

Code, protocols, simulation outputs and checkpoints are available at https://github.com/yeoneung/teacher_student_hj. Versioned artifacts and checksums are provided in release v1.0.0; Appendix M specifies the file layout.

Appendix A. Policy-gradient identity and transfer bounds

Appendix A.1. Teacher gradient

For a differentiable policy π_θ with teacher parameters θ_ϕ , freeze teacher states and covectors in

$$\mathcal{S}_\phi(\theta) = \mathbb{E} \sum_{t=0}^{H-1} \left[c_t(x_t^\phi, \pi_\theta(t, x_t^\phi)) + (p_{t+1}^\phi)^\top F_t(x_t^\phi, \pi_\theta(t, x_t^\phi)) \right]. \quad (\text{A.1})$$

When differentiation and expectation commute, reverse differentiation gives

$$\nabla_\theta \mathcal{S}_\phi|_{\theta_\phi} = \nabla_\theta J(\pi_\theta)|_{\theta_\phi} \quad (\text{A.2})$$

$$= \mathbb{E} \sum_t (D_\theta \pi_{\theta_\phi}(t, x_t^\phi))^\top [c_u + F_u^\top p_{t+1}^\phi]. \quad (\text{A.3})$$

Here $D_\theta \pi$ holds the state fixed; the covectors carry feedback effects. Every decision time contributes. The proximal term has zero gradient at the teacher. This identity holds at θ_ϕ ; repeated student updates can change both occupancy and covectors.

Appendix A.2. Transfer bounds

For a fixed teacher, set $u_0 = \phi_t(x)$, $y_0 = F_t(x, u_0)$ and $p = \nabla V_{t+1}^\phi(y_0)$. Assume the teaching sensitivity $\hat{p}$ satisfies

$$\|B_t^\top (\hat{p} - p)\| \leq \varepsilon, \quad (\text{A.4})$$

$$V_{t+1}^\phi(y_0 + B_t v) - V_{t+1}^\phi(y_0) - p^\top B_t v \leq \frac{\kappa}{2} \|v\|^2 \quad (\text{A.5})$$

on the relevant feasible segments. An L -Lipschitz teacher gradient permits $\kappa = L\|B_t\|^2$; the implementation does not know this bound.

Proposition 4. *Let $r > 0$, $\rho \geq 0$, and $\hat{q}(u) = r\|u\|^2 + (l + B_t^\top \hat{p})^\top u$. Let u_+ minimize $\hat{q}(u) + \rho\|u - u_0\|^2/2$ over the closed convex feasible set, and put $d = u_+ - u_0$. Then*

$$A_t^\phi(x, u_+) \leq -(r + \rho - \kappa/2)\|d\|^2 + \varepsilon\|d\|. \quad (\text{A.6})$$

For a fitted feasible action $u = u_+ + e$,

$$\begin{aligned} A_t^\phi(x, u) \leq & -(r + \rho)\|d\|^2 + \|\nabla \hat{q}(u_+)\| \|e\| + r\|e\|^2 \\ & + \varepsilon\|d + e\| + \frac{\kappa}{2}\|d + e\|^2 =: M_t^\phi(x, u). \end{aligned} \quad (\text{A.7})$$

If these assumptions hold on student trajectories with integrable bounds, $J(\pi) - J(\phi) \leq \mathbb{E} \sum_t M_t^\phi(x_t^\pi, \pi_t(x_t^\pi))$.

Proof. The proximal variational inequality and quadratic expansion give $\hat{q}(u_+) - \hat{q}(u_0) \leq -(r + \rho)\|d\|^2$. Expansion at u_+ bounds fitting by $\|\nabla\hat{q}(u_+)\| \|e\| + r\|e\|^2$. Sensitivity error and continuation remainder contribute at most $\varepsilon\|u - u_0\|$ and $\kappa\|u - u_0\|^2/2$. Summing along student trajectories proves the policy bound. $\square$

For $r + \rho - \kappa/2 > 0$, the sufficient target test is $(r + \rho - \kappa/2)\|d\| > \varepsilon$. With $\hat{g} = 2ru_0 + l + B_t^\top \hat{p}$ and $\kappa \geq 0$, nonexpansivity gives

$$\|d\| \leq \frac{\|\hat{g}\|}{2r + \rho}. \quad (\text{A.8})$$

Hence $\varepsilon \geq \|\hat{g}\|$ prevents every nonzero step from passing this test for any ρ . This limits the certificate, not the possibility of actual improvement.

Appendix A.3. Continuation accounting

Define $G = \hat{q}(u_0) - \hat{q}(u_+)$ and the signed fitting contribution $R = \hat{q}(u) - \hat{q}(u_+)$. Then

$$A_t^\phi(x, u) = -G + R + [B_t^\top (p - \hat{p})]^\top (u - u_0) + C, \quad (\text{A.9})$$

where C is the exact teacher-value remainder. Using a cached covector from another state contributes to the sensitivity discrepancy. A teacher continuation after the tested action evaluates the advantage exactly in the known simulator. Measured remainders verify this identity but do not supply prospective curvature bounds. The audit procedure uses sampled advantages to choose fitting, refresh or regularization; it does not estimate certified ε or κ .

Appendix B. Complete grid results

All 480 fits complete without nonfinite updates. Recovery and query-count checks pass in all 80 paired cache conditions. Maximum coefficient error is 2.49×10^{-15} ; relevant/full normalized-regret difference is at most 3.81×10^{-12} , meeting the tolerance in all 80 pairs. Maximum parameter difference is 8.94×10^{-7} . Re-evaluation reproduces every scalar metric; telescoping error is at most 7.74×10^{-14} .

Box caches require 332,362 relevant responses versus 1,039,456 for full-face recovery, a 68.03% reduction. At $d = 32, m = 2, R = 0.5$ the reduction is 85.20%, or 0.995 versus 6.722 queries per state including inactive targets. Table B.3 gives all box conditions.

Table B.3: Scalar feedback in the box experiment. Queries are means per cached state, including interior targets. Savings compare relevant queries with full normal-space reconstruction. Coefficient errors are relative RMS errors averaged over five seeds; these are information diagnostics, not policy costs.

d, m, R	Relevant	Full face	Saved (%)	Gain error	Random error
8,2,0.5	0.876	1.592	45.0	0.108	0.184
8,2,2	0.073	0.073	0.0	0.000	0.000
8,4,0.5	1.388	1.592	12.8	0.107	0.107
8,4,2	0.073	0.073	0.0	0.000	0.000
32,2,0.5	0.995	6.722	85.2	0.194	0.379
32,2,2	0.072	0.072	0.0	0.000	0.000
32,4,0.5	1.861	6.722	72.3	0.178	0.326
32,4,2	0.072	0.072	0.0	0.000	0.000

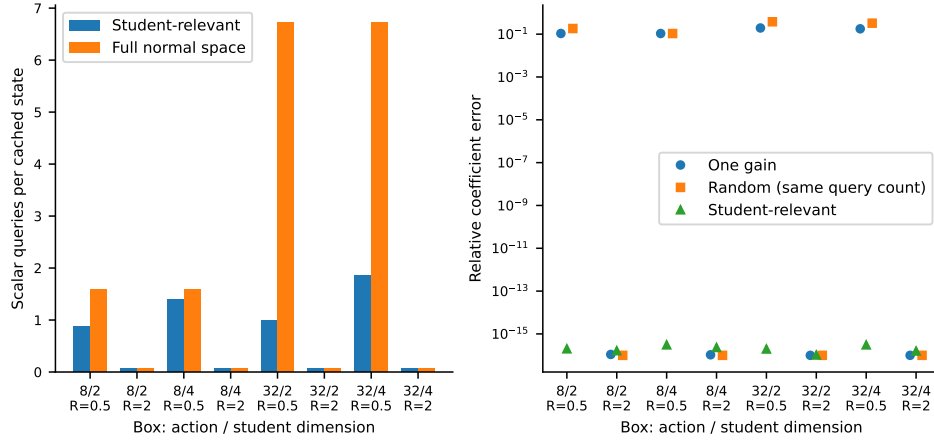


Figure B.5: Box query counts and relative coefficient errors, averaged over five seeds. Relevant and random methods use the same number of responses; gain uses one per active target.

On balls, all four scalar recovery methods reproduce full-information learning. Sampled boxes at $R = 2$ have at most one active coordinate, so gain and random also recover exactly and no query saving occurs. At $R = 0.5$, their condition-mean coefficient errors range from 0.107 to 0.379. Policy effects are smaller: for $d = 32, m = 2, R = 0.5$, relevant, gain and random regrets are 0.90062, 0.90285 and 0.90632.

Relevant feedback lowers mean regret and cost versus target-only in all 16 conditions. For each endpoint, three unadjusted intervals include zero: both eight-dimensional ball conditions with $m = 4$, and the eight-dimensional box with $m = 4, R = 2$ (Appendix C). High remaining regret in small subspaces reflects limited representability.

Algebraic checks cover 576 recovery cases through dimension 128, including a nonzero student-orthogonal normal needing no query. Another 120 constructions use $r - 1$ queries: admissible normals yield identical responses within 6.40×10^{-17} but different student coefficients. They also verify Eq. (25) for response error norms up to 10^{-3} . These checks involve no additional training.

Appendix C. Complete minimal-feedback results

Each entry reports five independent seed means; all methods share the initialization, data and 4,096-update budget within a seed. The full-face method is retained in the machine-readable results and the fidelity audit. Intervals below are paired 95% t intervals for relevant-query minus target-only regret, with no adjustment across the 16 secondary comparisons. Recovery, not superiority in this table, is the primary endpoint.

Table C.4: Normalized local Bellman regret for every prespecified condition. Smaller is better. Relevant uses the minimum scalar-query construction; full uses privileged exact normal vectors.

Shape; d, m, R	Target	Gain	Random	Relevant	Full
ball; 8,2,0.5	0.69427	0.59141	0.59141	0.59141	0.59141
ball; 8,2,2	0.71503	0.65222	0.65222	0.65222	0.65222
ball; 8,4,0.5	0.38279	0.34317	0.34317	0.34317	0.34317
ball; 8,4,2	0.41671	0.39429	0.39429	0.39429	0.39429
ball; 32,2,0.5	0.93161	0.84013	0.84013	0.84013	0.84013
ball; 32,2,2	0.93528	0.90773	0.90773	0.90773	0.90773
ball; 32,4,0.5	0.85415	0.71537	0.71537	0.71537	0.71537
ball; 32,4,2	0.85541	0.80166	0.80166	0.80166	0.80166
box; 8,2,0.5	0.71992	0.58590	0.58641	0.58544	0.58544
box; 8,2,2	0.71504	0.59140	0.59140	0.59140	0.59140
box; 8,4,0.5	0.45383	0.38937	0.38900	0.38887	0.38887
box; 8,4,2	0.39893	0.34412	0.34412	0.34412	0.34412
box; 32,2,0.5	0.93335	0.90285	0.90632	0.90062	0.90062
box; 32,2,2	0.93947	0.92451	0.92451	0.92451	0.92451
box; 32,4,0.5	0.86405	0.81009	0.81197	0.80780	0.80780
box; 32,4,2	0.86617	0.82033	0.82033	0.82033	0.82033

Table C.5: All paired relevant-query minus target-only differences. Cost is the complete 12-step closed-loop cost, whereas regret is measured on the held-out teacher-state pool.

Shape; d, m, R	Regret difference [95% CI]	Cost difference [95% CI]
ball; 8,2,0.5	-0.10286 [-0.12893, -0.07680]	-0.08206 [-0.11111, -0.05302]
ball; 8,2,2	-0.06282 [-0.08203, -0.04361]	-0.11445 [-0.15633, -0.07258]
ball; 8,4,0.5	-0.03962 [-0.07955, +0.00031]	-0.03757 [-0.08616, +0.01102]
ball; 8,4,2	-0.02242 [-0.06062, +0.01578]	-0.03332 [-0.10633, +0.03969]
ball; 32,2,0.5	-0.09148 [-0.10093, -0.08203]	-0.10667 [-0.12492, -0.08842]
ball; 32,2,2	-0.02754 [-0.03663, -0.01845]	-0.04519 [-0.07492, -0.01546]
ball; 32,4,0.5	-0.13878 [-0.14871, -0.12885]	-0.17655 [-0.19985, -0.15324]
ball; 32,4,2	-0.05376 [-0.06418, -0.04334]	-0.10600 [-0.14875, -0.06325]
box; 8,2,0.5	-0.13448 [-0.17990, -0.08906]	-0.17735 [-0.23849, -0.11620]
box; 8,2,2	-0.12364 [-0.17123, -0.07605]	-0.21152 [-0.29792, -0.12512]
box; 8,4,0.5	-0.06495 [-0.10745, -0.02246]	-0.10366 [-0.16229, -0.04503]
box; 8,4,2	-0.05481 [-0.11787, +0.00825]	-0.09722 [-0.21235, +0.01791]
box; 32,2,0.5	-0.03273 [-0.04732, -0.01815]	-0.09151 [-0.14473, -0.03828]
box; 32,2,2	-0.01496 [-0.02663, -0.00328]	-0.05216 [-0.09303, -0.01130]
box; 32,4,0.5	-0.05624 [-0.07451, -0.03798]	-0.20451 [-0.28722, -0.12179]
box; 32,4,2	-0.04584 [-0.06910, -0.02259]	-0.17173 [-0.26299, -0.08047]

Appendix D. Alignment and response-error checks

For the normal space $K = \text{span}(e_1, e_2, e_3)$, set $g = (e_1 + e_2 + e_3)/\sqrt{3}$, $h_1 = (e_1 - e_2)/\sqrt{2}$ and $h_2 = (e_1 + e_2 - 2e_3)/\sqrt{6}$. The perturbed embedding is

$$P_\theta = [g, \cos \theta e_4 + \sin \theta h_1, \cos \theta e_5 + \sin \theta h_2, e_6]H. \quad (\text{D.1})$$

Its nonzero candidate singular values are $(1, \sin \theta, \sin \theta)$. Exact rank is one at zero and three for every tested $\theta > 0$. Angles are 0, 10^{-6} , 10^{-4} , 0.01, 0.1 and 0.5 radians. We use one or three queries, response-error norms 0, 10^{-6} , 10^{-3} , and 64 constructed normals per cell. Negative normals of norm one on the first three coordinates, with target $(0.5, 0.5, 0.5, 0, \dots, 0)$, define compatible quadratics through $b = n - 2av$. Feasible queries use $\alpha = 0.25$.

All 2,304 checks satisfy Eq. (26) and its cost-difference bound. For one noiseless query the bound is $\sin \theta$; response noise adds orthogonally. Figure D.6 shows the observed maxima. These are algebraic checks with known $M = 1$, not noisy-network training or evidence that arbitrary perturbations preserve the one-query exact result. The neural implementation’s Gram eigenvalue threshold 10^{-10} would discard sufficiently small positive singular values; the perturbed checks use the explicit spectrum above.

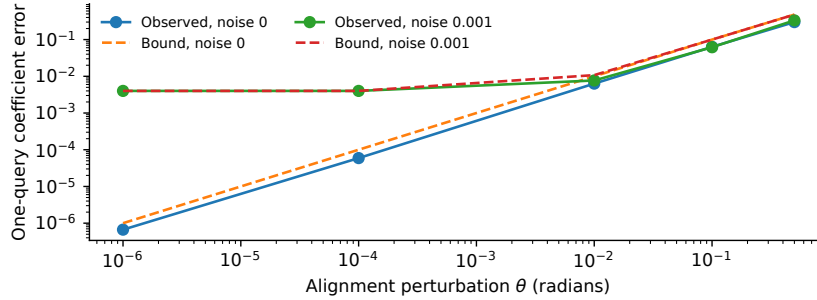


Figure D.6: One-query error under perturbed alignment. Points show maxima over 64 compatible normals per cell; dashed curves give the upper bound.

Appendix E. Improvement sets

An advantage sublevel set allows the student to choose among actions meeting a specified improvement requirement.

Appendix E.1. Construction

At $z = (t, x)$, let $u_0 = \phi_t(x)$, $y_0 = F_t(x, u_0)$, $p = \nabla V_{t+1}^\phi(y_0)$, and $\hat{q}_z(u) = r_t \|u\|^2 + (l_t + B_t^\top \hat{p})^\top u$. Assume compact convex nonempty U , $u_0 \in U$, $r_t > 0$, and, for all $u \in U$,

$$\|B_t^\top(\hat{p} - p)\| \leq \varepsilon_z, \quad (\text{E.1})$$

$$V_{t+1}^\phi(y_0 + B_t(u - u_0)) - V_{t+1}^\phi(y_0) - p^\top B_t(u - u_0) \leq \frac{\kappa_z}{2} \|u - u_0\|^2, \quad (\text{E.2})$$

with $\varepsilon_z, \kappa_z \geq 0$. The upper model

$$\mathcal{B}_z(u) = \hat{q}_z(u) - \hat{q}_z(u_0) + \varepsilon_z \|u - u_0\| + \frac{\kappa_z}{2} \|u - u_0\|^2 \quad (\text{E.3})$$

satisfies $A_z^\phi(u) \leq \mathcal{B}_z(u)$ and $\mathcal{B}_z(u_0) = 0$. For a feasible candidate w_z , define the realized anchor and gain

$$a_z = \begin{cases} w_z, & \mathcal{B}_z(w_z) < 0, \\ u_0, & \text{otherwise,} \end{cases} \quad \hat{g}_z = -\mathcal{B}_z(a_z) \geq 0. \quad (\text{E.4})$$

Use this same anchor for point and set supervision. For $0 < \alpha \leq 1$,

$$C_\alpha(z) = \{u \in U : \mathcal{B}_z(u) \leq -\alpha \hat{g}_z\}. \quad (\text{E.5})$$

Proposition 5. *The set $C_\alpha(z)$ is nonempty, compact and convex, contains a_z , and every member satisfies $A_z^\phi(u) \leq -\alpha \hat{g}_z$. Its projection is unique and nonexpansive, and*

$$\nabla_u \text{dist}(u, C_\alpha(z))^2 = 2\{u - \text{proj}_{C_\alpha(z)} u\}. \quad (\text{E.6})$$

Proof. Continuity and strong convexity of $\mathcal{B}_z$, compactness of U , and $-\hat{g}_z \leq -\alpha \hat{g}_z$ give the set properties. The advantage inequality follows from its definition; the projection claims follow from the convex projection variational inequality. $\square$

At $\alpha = 1$ the set is a singleton only if the anchor minimizes $\mathcal{B}_z$. When $\hat{g}_z = 0$, it still permits modeled non-worsening actions.

Appendix E.2. Cost bound

For a distribution μ and feasible policy class Π ,

$$L_{\text{set}}(\pi) = \mathbb{E}_\mu \text{dist}(\pi(z), C_\alpha(z))^2 \leq \mathbb{E}_\mu \|\pi(z) - a_z\|^2 = L_{\text{point}}(\pi). \quad (\text{E.7})$$

Lower set loss need not mean lower cost. For equiprobable costs $(u-1)^2, (u-2)^2$, $U = [-3, 3]$, teacher zero and constant students, the best point fit is 1.5 with MSE 0.25 and cost 0.25. At $\alpha = 1/2$, constant 1 has zero set loss but cost 0.5; teacher cost is 2.5.

Proposition 6. Suppose the upper-model assumptions hold on trajectories of a feasible student π . Let L_z be a Lipschitz constant of $\mathcal{B}_z$ on U and $\delta_z = \text{dist}(\pi(z), C_\alpha(z))$. With finite expectations,

$$\begin{aligned} J(\pi) - J(\phi) &\leq -\alpha \mathbb{E} \sum_t \hat{g}_{z_t} + \mathbb{E} \sum_t L_{z_t} \delta_{z_t} \\ &\leq -\alpha \mathbb{E} \sum_t \hat{g}_{z_t} + \sqrt{\mathbb{E} \sum_t L_{z_t}^2} \sqrt{\mathbb{E} \sum_t \delta_{z_t}^2}, \quad z_t = (t, x_t^\pi). \end{aligned} \quad (\text{E.8})$$

An admissible constant is

$$L_z = \sup_{u \in U} \|2r_t u + l_t + B_t^\top \hat{p}\| + \varepsilon_z + \kappa_z \text{diam}(U). \quad (\text{E.9})$$

Proof. For $v = \text{proj}_{C_\alpha(z)} \pi(z)$, $A_z^\phi(\pi(z)) \leq \mathcal{B}_z(v) + L_z \|\pi(z) - v\| \leq -\alpha \hat{g}_z + L_z \delta_z$. Sum along student trajectories and apply Cauchy–Schwarz. The displayed L_z bounds the three terms of Eq. (E.3). $\square$

An upper-model error ξ_z adds $\mathbb{E} \sum_t \xi_{z_t}$. Cache loss requires a coverage argument to control this student-occupancy bound.

Appendix E.3. Projection loss

For $\varepsilon_z = 0$ and a Euclidean input ball of radius $R_U = \sqrt{n} \bar{u}$, set $a = r_t + \kappa_z/2$, $h_z = 2r_t u_0 + l_t + B_t^\top p$ and $m_z = u_0 - h_z/(2a)$. Then

$$C_\alpha(z) = U \cap \{u : \|u - m_z\| \leq R_z\}, \quad R_z^2 = \frac{\|h_z\|^2/(4a) - \alpha \hat{g}_z}{a}. \quad (\text{E.10})$$

Projection uses single-ball projections or their boundary intersection, with linear arithmetic cost in action dimension (Appendix N). This formula is restricted to the stated geometry and isotropic costs. The implementation caches set parameters, recomputes projection at every update, and detaches it when differentiating squared distance, as justified by Eq. (E.6). The teacher stays fixed; deployment uses only the student. Unlike policy-update projection [16], this projection defines action-level supervision.

Appendix E.4. Joint realizability

Define the set of students that satisfy every teaching set on a specified collection Z :

$$\Pi_\alpha(Z) = \{\pi \in \Pi : \pi(z) \in C_\alpha(z) \text{ for every } z \in Z\}. \quad (\text{E.11})$$

Proposition 7. *For the exact model in Proposition 2, with $\tau = 10$ and the exact minimizer anchors, every $C_\alpha(i)$ is nonempty for $0 < \alpha \leq 1$, but no constant student satisfies all three sets. Nevertheless a constant student strictly improves expected teacher cost. At $\alpha = 1/2$, even the global minimum of mean squared set distance has lower set loss and higher cost than the teacher.*

Proof. Here $\mathcal{B}_i(u) = Q_i(u) - Q_i(0)$ and the anchor gains are $(19, 1, 1)$. Solving their sublevel inequalities on $[-1, 1]$ gives

$$C_\alpha(1) = [10 - \sqrt{100 - 19\alpha}, 1], \quad (\text{E.12})$$

$$C_\alpha(2) = C_\alpha(3) = [-1, -1 + \sqrt{1 - \alpha}]. \quad (\text{E.13})$$

The first lower endpoint is strictly positive and the other upper endpoint strictly negative for every $\alpha > 0$. Their intersection is empty. The constant $c = 1$ nevertheless has cost $89/3 < 34$: improvement in the first context compensates for worsening in the others. Write $l = 10 - \sqrt{90.5}$ and $h = -1 + 1/\sqrt{2}$ for the endpoints at $\alpha = 1/2$. Between h and l , the mean squared set distance is $[(c - l)^2 + 2(c - h)^2]/3$. Convexity gives the global minimizer $c = (l + 2h)/3 \approx -0.03297841$. Since $l + 2h < 0$, this constant is negative, so its cost difference $c^2 - 16c/3$ is strictly positive. Its set loss decreases by $c^2 > 0$. Numerically its set loss is 0.135111, below the teacher’s 0.136199, while its cost is 34.176972, above 34. $\square$

Pointwise improvement can therefore impose jointly infeasible requirements even when average-cost improvement is attainable.

Appendix F. Continuation-audited teaching

The nonlinear diagnostic procedure keeps teacher ϕ fixed and maintains accepted student ψ and trainable proposal π_θ . The initial student is a weaker pretrained policy of the same architecture.

Appendix F.1. Updates and audit rule

The initial cache contains 4,096 state/time pairs sampled with replacement from initial-student trajectories, storing features, base-controller outputs and projected teacher covectors. Proximal targets are fitted with normalized action MSE, Adam learning rate 0.002 and batch 512. The coefficient $\eta = \rho/(2r)$ starts at 16. Blocks contain 128 updates, with timing checks every 16.

After each block, sample 16 state/time pairs from proposal trajectories on four fixed training initials. Let $\bar{A}_+$ and $\bar{A}_\pi$ be mean exact teacher advantages of fresh targets and proposal actions:

$$\text{operation} = \begin{cases} \text{regularize,} & \bar{A}_+ \geq -10^{-7}, \\ \text{fit,} & \bar{A}_+ < -10^{-7} \text{ and } \bar{A}_\pi > \bar{A}_+/2, \\ \text{refresh,} & \text{otherwise.} \end{cases} \quad (\text{F.1})$$

Regularization doubles η up to 4,096. Fitting preserves proposal and Adam state below 1,024 updates since the last reset. Refresh replaces a 64-entry cache window using proposal trajectories on eight training initials; the other 4,032 initial entries stay fixed.

A target using the nearest cached covector, under RMS state distance plus $10|t - t'|/H$, is also evaluated. Its cost is charged and logged but does not change the branch. The rule does not separately identify sensitivity, curvature and coverage errors.

Appendix F.2. Acceptance and budget

Each audit evaluates full proposal cost on 512 training initials. Accept finite cost within $10^{-7}(1 + |J_{\text{incumbent}}|)$ of the incumbent. Acceptance updates only the student. Accepted proposals are evaluated on 256 validation initials and become eligible if evaluation and copying finish within budget. Return the eligible policy with lowest validation cost, using initialization as fallback.

The fitting branch may retain a rejected proposal internally; it remains ineligible until accepted. Otherwise rejection restores the incumbent. Adam resets after acceptance or rejection unless fitting carries forward.

The 30-second teaching budget includes graph/cache preparation, queries, continuations, fitting, rejections, validation and synchronized copies. Teacher and initial-student pretraining have separate 30- and 7.5-second budgets. Common process/CUDA initialization and final disk writing are excluded; late checkpoints cannot be selected. Reverse sweeps avoid a state-space grid but retain horizon-dependent storage and audit rollouts.

Appendix G. Audit experiment protocol

Appendix G.1. Tasks and policies

The synthetic known-model tasks are a mechanical ring with 16 nodes, 32 states and horizon 160, and a reaction ring with 32 states and horizon 320. Both use control-affine dynamics, quadratic input costs and RMS constraints

(Appendix J). One timed experiment runs at a time on an RTX 4080 SUPER, PyTorch 2.7.1 and CUDA 11.8.

All policies use a node-shared residual actor with two width-64 SiLU layers and time, local, neighboring and pooled state features. A bounded residual is added to a projected linearized-LQR base, then projected to the input set. Mechanical teachers are 30-second Short policies trained with 16-step rollouts and the stated terminal penalty; reaction teachers are 30-second Warm DPC policies using full-horizon derivatives. Initial students are 7.5-second Short policies. These pretrained inputs are separate from the teaching runs.

Appendix G.2. Comparators and evaluation

Five paired teacher/student/data replications per task compare cloning, cached Hamiltonian fitting, fixed targets, adaptive targets and audited teaching. Within a replicate they share the teacher, initial student, cache, architecture, 512 training initials, 256 validation initials and 30-second allocation. After saving each selected policy, generate 512 separate test initials shared across methods and teacher.

Cloning fits teacher actions. Cached Hamiltonian fitting uses the normalized proximal quadratic with $\eta = 16$. Fixed targets also use 16, selected from $\{16, 64, 256\}$ in a separate 15-second validation experiment. Adaptive fitting doubles η on rejection, invalid predicted reduction or an actual-to-predicted reduction ratio below 0.25; it halves η above 0.75, within $[0.25, 4096]$. Teacher and cache remain fixed.

Baselines gate acceptance and reset Adam every 512 updates; audited teaching checks every 128 and may retain fitting state. Thus schedules, coverage and retention change jointly. Common covector preparation is charged even to cloning. Report seed-mean test costs and paired 95% t intervals with four degrees of freedom, without multiplicity adjustment, test-based exclusion or retuning.

Appendix G.3. Diagnostics

Each selected student supplies 32 state/time pairs from four independent diagnostic trajectories. Fresh covectors and teacher continuations evaluate Eq. (A.9); all post-fit targets use $\eta = 16$. The 160 pairs per task are dependent samples and differ from training targets and complete test-trajectory sums.

An action-level experiment crosses five teacher seeds, checkpoint budgets 7.5/15/30 seconds, and five sensitivity conditions: unperturbed, fixed bias or Gaussian noise at relative magnitudes 0.5/1.5. Each task/seed

reuses 32 weak-student state/time pairs. The per-coordinate scale is $\max(\text{RMS}(B^\top p), 2r\bar{u})$. Injected norms are known; no student is retrained.

Curvature is calibrated independently as 1.5 times the 95th percentile of positive directional-remainder coefficients on two complete trajectories per task (Appendix K): 0.000550492 and 0.000417389. These fixed empirical values are assessed by violations, false positives and coverage. Improvement/worsening thresholds are $-10^{-8}/10^{-8}$; violation tolerance is 10^{-7} .

Appendix H. Audit results

Appendix H.1. Policy cost

Audited teaching lowers initial-student cost by 7.74% in mechanics and 18.64% in reaction, with five wins per task, but remains 0.42% and 2.60% above the teacher (Table H.6). Student-minus-teacher intervals are $[-0.012495, 0.036137]$ and $[-0.021056, 0.058030]$; one mechanical and three reaction students beat their teachers. Versus cached Hamiltonian fitting,

Task	Method	Cost	Updates	Audits
Mechanical	Teacher	2.814376	–	–
Mechanical	Initial student	3.063236	–	–
Mechanical	Cloning	2.826377	28938	0.0
Mechanical	Cached H	2.864541	29770	0.0
Mechanical	Selected fixed	2.825197	29674	0.0
Mechanical	Adaptive	2.829827	29766	0.0
Mechanical	Audit-guided	2.826198	4019	31.4
Reaction	Teacher	0.711506	–	–
Reaction	Initial student	0.897189	–	–
Reaction	Cloning	0.745016	21357	0.0
Reaction	Cached H	0.822687	21507	0.0
Reaction	Selected fixed	0.835997	21731	0.0
Reaction	Adaptive	0.731010	21510	0.0
Reaction	Audit-guided	0.729993	3232	25.2

Table H.6: Mean costs and computation under the common 30-second teaching allocation, using five paired replications per task. Teacher and initial student policies are pretrained inputs; their construction costs are separate. All continuation audits and common preparation are charged.

mean costs fall 1.34% and 11.27%, with five wins and paired intervals $[-0.073294, -0.003394]$ and $[-0.161527, -0.023861]$. Against fixed targets, changes are +0.0354% and −12.68%; against adaptive targets, −0.128% and −0.139%. These and cloning intervals include zero (Appendix L). The

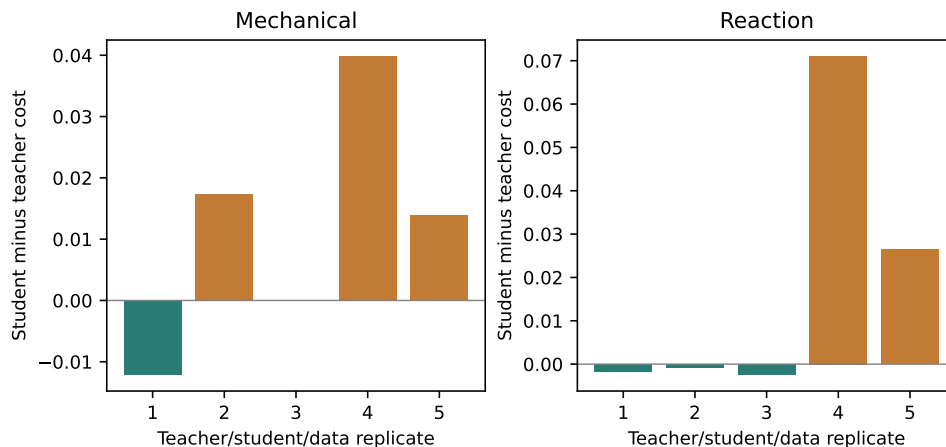


Figure H.7: Cost relative to each replicate’s teaching policy. Negative bars indicate a student that improves on its teacher. All runs are retained.

comparison therefore supports a gain over the specified cached-Hamiltonian configuration, with no established benefit over adaptive regularization.

Appendix H.2. Local diagnostics

With fresh exact sensitivities and common $\eta = 16$, the positive fitting term exceeds the target’s negative model change in both tasks. Mean student advantage remains positive (Table H.7). Representation, optimization,

Mean term	Mechanical	Reaction
Target-model change $-G$	-1.373805×10^{-2}	-2.214686×10^{-5}
Fitting contribution R	$+1.415783 \times 10^{-2}$	$+3.959945 \times 10^{-5}$
Nonlinear remainder C	$+3.066642 \times 10^{-4}$	$+1.137840 \times 10^{-7}$
Exact student advantage	$+7.264437 \times 10^{-4}$	$+1.756637 \times 10^{-5}$

Table H.7: Exact accounting on 160 sampled student state/time pairs per task. The diagnostic uses a common target coefficient and is separate from the training-time target sequence. Terms are signed averages, not nonnegative losses.

training-target differences and coverage can all contribute; the probes do not isolate them. The fresh local model has two false positives among 101 predicted mechanical improvements and none among 33 reaction predictions. Among 3,840 evaluations with nonzero injected noise, the quadratic model predicts improvement in every case and is wrong in 2,993. The empirical margin predicts 436 improvements with no false positives, abstaining 88.65%

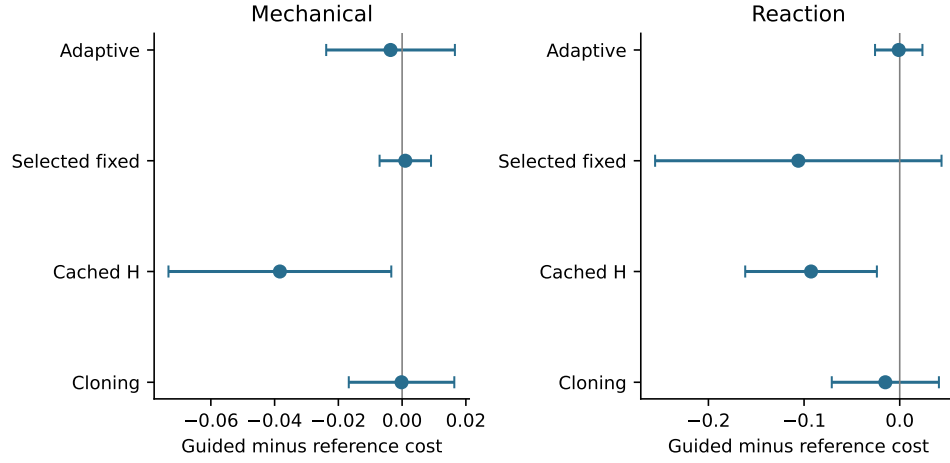


Figure H.8: Audited-student minus teaching-baseline cost, with exploratory 95% paired t intervals. The adaptive baseline changes the proximal coefficient inside the same fixed-teacher setting.

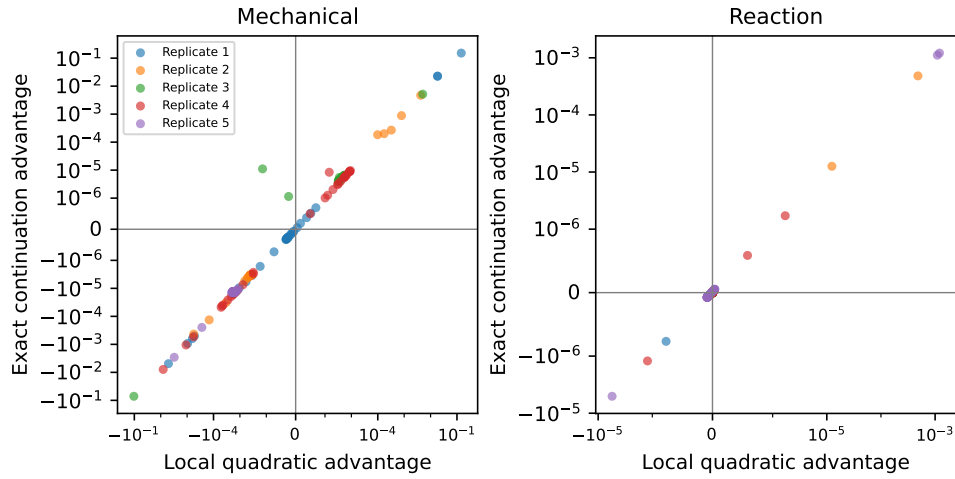


Figure H.9: Local quadratic versus exact continuation advantage for the selected audited students. Symmetric-log axes retain positive and negative values. The upper-left quadrant contains false positive local predictions.

of the time. At reaction Gaussian magnitude 1.5 it abstains even on four genuinely improving targets. States are reused across conditions. Across all 4,800 evaluations there are 26 curvature-bound violations (12 mechanical, 14 reaction), and two unperturbed mechanical false-positive margin predictions (Appendix K).

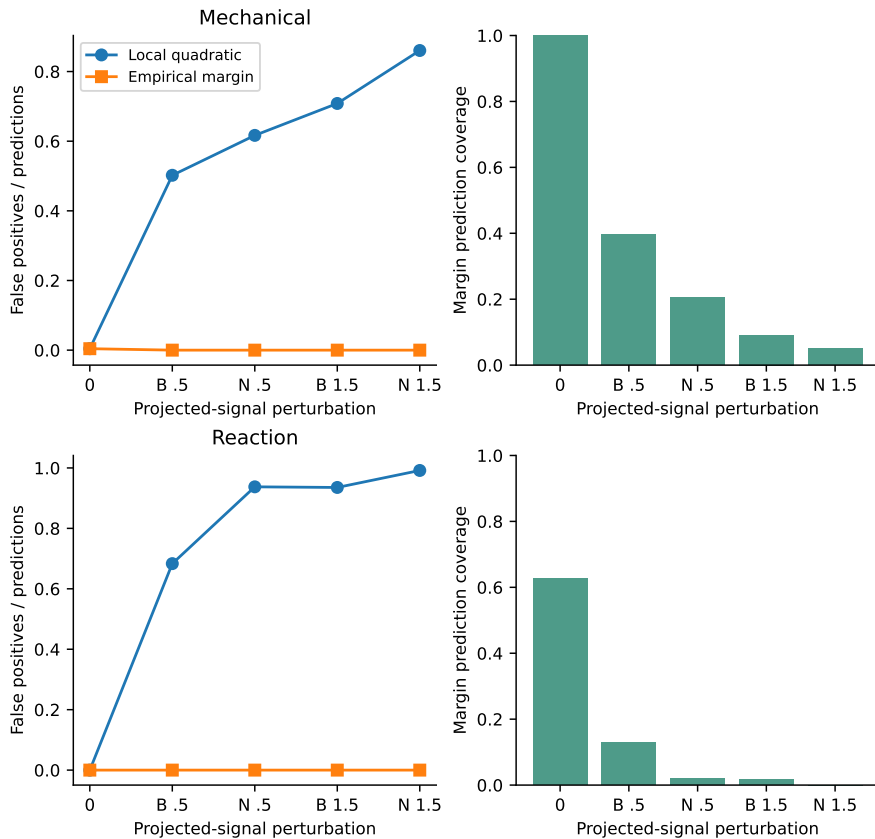


Figure H.10: False positives and coverage under projected-sensitivity perturbations. B denotes fixed bias and N Gaussian noise. Each task/condition has 480 action evaluations from 160 sampled pairs across three teacher qualities. Zero predictions in the final reaction condition denote abstention.

Appendix H.3. Computation

Mechanical runs request 107 fitting extensions and 50 refreshes; reaction runs request 66 extensions, 16 refreshes and 44 regularizations. Common preparation averages 11.54 and 14.07 seconds (38.5% and 46.9% of budget). Audited teaching averages 4,019 and 3,232 updates, versus 29,770 and 21,507

for cached Hamiltonian fitting. Continuations, the unused cached-target probe and the small refresh window impose overhead without isolating each branch’s contribution to final cost.

Appendix I. Improvement-set experiments

Appendix I.1. Protocol

This matched-update comparison shares teacher, initial student, pool, optimizer, refresh and cost gate, without the audit branches. Point fitting regresses the realized anchor; set fitting recomputes projection onto C_α at every update; upper-model fitting minimizes $\mathcal{B}_z$ normalized by its quadratic coefficient. Point and set use $\eta = 16$. Adaptive point doubles η on rejection, invalid prediction or reduction ratio below 0.25, and halves it above 0.75, within $[0.25, 4096]$.

The teacher is a fixed 30-second checkpoint. Runs use 64 training, 128 validation and 512 disjoint test initials, generated after selected checkpoint saving; initial cache 512, batch 256, Adam learning rate 0.002 and 1,024 updates. Every 256 updates a training-cost gate rejects or accepts; rejection restores accepted parameters and Adam always resets. After update 512, replace half the cache with 256 pairs from accepted student trajectories on eight training initials. Validation selects among accepted checkpoints and initialization. Hidden weights and minibatch seeds are paired within architectures. Sensitivities are exact, $\varepsilon = 0$, with empirical $\kappa = 0.000550492$ in mechanics and 0.000417389 in reaction.

Exploration uses both tasks, widths 16/64, $\alpha \in \{0.25, 0.5, 0.75\}$, and zero residual over the analytic base (24 fits), plus pretrained width-64 students from the 7.5-second parent (12 fits). Each condition compares point, adaptive point, upper model and three set relaxations with one teacher/data seed per task (Appendix O). Only pretrained reaction meets the prespecified replication criterion: at least 1% validation improvement over both point comparators, selecting $\alpha = 0.25$. Five independent teacher/student/data pairs then compare this relaxation with the three controls. Pilot samples are not pooled; paired 95% t intervals use five seed means without multiplicity adjustment. Charged time includes graph preparation, queries, projection, fitting, refresh, gates and validation, excluding pretraining; wall budgets are not matched.

Appendix I.2. Results

Exact projection and eight-dimensional quadratic checks are in Appendix N. Replication gives set cost 0.902697 versus point/adaptive

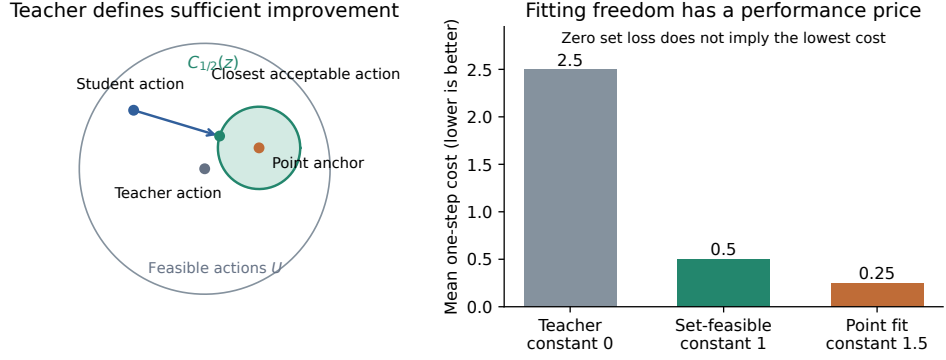


Figure I.11: The teacher supplies an acceptable-improvement set; the student is pulled toward its closest member. The right panel is the analytic two-state counterexample, where zero set loss permits a policy with higher cost than point fitting. These are geometric and analytic illustrations, not neural-training results.

Table I.8: Reaction-task improvement-set replication: five paired seed means after 1,024 updates. Times charge preparation and all teaching operations, with pretraining separate. The selected initial count includes both gate and validation effects.

Method	Mean test cost	Charged s	Initial selected
Point anchor	0.907593	6.524	4/5
Adaptive point	0.907593	6.517	4/5
Upper model	0.891263	6.523	4/5
Set, $\alpha = 0.25$	0.902697	6.570	3/5
Initial student	0.908277	—	—
Teacher	0.737725	—	—

0.907593 and upper-model 0.891263. Set-minus-point is -0.004896 $[-0.041095, 0.031303]$, with one win, one loss and three ties; set-minus-upper is 0.011434 $[-0.008016, 0.030885]$. Set cost remains 22.36% above teacher, with no teacher wins. Initialization is selected in 3/5 set runs

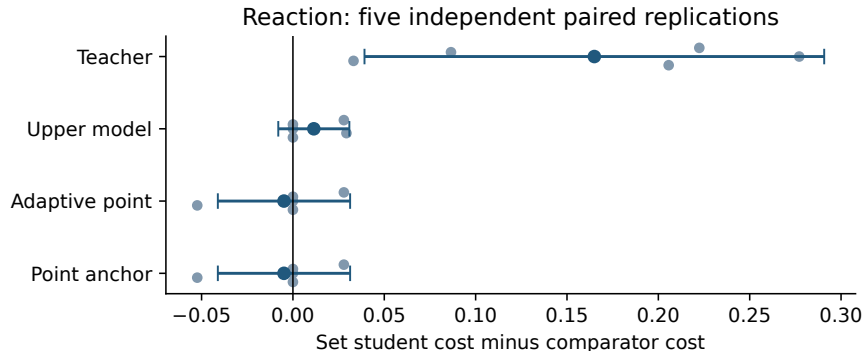


Figure I.12: Set-student cost minus comparator cost in independent reaction replications. Small points are the five paired differences; large points and bars give their means and unadjusted 95% t intervals. Negative values favor the set student.

and 4/5 comparator runs. The pretrained pilot lowers validation cost from 1.049305 to 0.938238 but raises test cost from 0.854282 to 0.993972. Analytic-base conditions give no selected set-policy improvement. All adverse outcomes are retained.

Post-fit probes use 64 pairs per student on four separate trajectories, with fresh $\eta = 16$ anchors and $\alpha = 0.25$. On set-student states, set MSE is 1.36737×10^{-4} versus point MSE 9.54646×10^{-4} ; point-trained students have set MSE 1.28877×10^{-4} on their own states. Membership is 37.5% versus 17.19%, but mean local advantages are 1.4860×10^{-6} versus -4.8371×10^{-6} . Each objective has one upper-model violation and one non-improving modeled-set member among 320 dependent sampled pairs. Thus the empirical sets are not certified. Charged time averages 6.570 seconds for set and 6.524 for point fitting; the fitting portions are 0.427 and 0.376 seconds.

Appendix J. Dynamics, costs and policy inputs

All ring indices are periodic, and $\langle z \rangle = n^{-1} \sum_i z_i$. The mechanical system uses $h = 0.08$, coupling 2.5, RMS control bound 3.2, and

$$v_i^+ = v_i + h[5 \sin q_i + 2.5\{\sin(q_{i-1} - q_i) + \sin(q_{i+1} - q_i)\} + u_i - 0.2v_i], \quad (\text{J.1})$$

$$q_i^+ = q_i + hv_i^+, \quad (\text{J.2})$$

$$c(x, u) = h\langle 2(1 - \cos q_i) + 0.08v_i^2 + 0.04u_i^2 + 0.2(1 - \cos(q_i - q_{i+1})) \rangle, \quad (\text{J.3})$$

$$g(x) = \langle 8(1 - \cos q_i) + 0.4v_i^2 \rangle. \quad (\text{J.4})$$

Initial positions are $2.2z_i + 0.6z_0$ and velocities $1.5w_i$, where the variables are independent uniform $[-1, 1]$ except the common z_0 . The reaction system uses $h = 0.05$, coupling 2 and RMS bound 0.8:

$$x_i^+ = x_i + h[2(x_{i-1} + x_{i+1} - 2x_i) + 1.2x_i - 0.4x_i^3 + u_i], \quad (\text{J.5})$$

$$c(x, u) = h\langle x_i^2 + 0.1u_i^2 + 0.1(x_i - x_{i+1})^2 \rangle, \quad g(x) = 3\langle x_i^2 \rangle. \quad (\text{J.6})$$

Initial values are $1.4z_i + z_0$.

Appendix K. Diagnostic calibration

The independent curvature calibration uses two complete student trajectories per task. The teaching policies are 7.5-second Short checkpoints. Calibration students are unaudited proximal-target policies: 256-update fitting cycles, full training-cost acceptance, teacher promotion on acceptance, and an agreement-ratio coefficient schedule starting at 16. They use a 30-second total allocation and separate teacher seed 9501. These policies supply calibration trajectories only. Reverse differentiation evaluates each fixed teacher's value gradient at the next state after its own action. The positive directional-remainder coefficient is the larger of $2[C]_+/\|u - u_0\|^2$ for the tested student action and proximal target, with a 10^{-20} denominator floor. Take its 95th percentile and multiply by 1.5 for the fixed empirical parameter. This coefficient uses realized remainders and is not a prospective bound.

On these complete calibration trajectories, the maximum difference between summed exact teacher advantages and the independently evaluated policy cost difference is 3.94×10^{-7} .

The perturbation experiment uses 150 task/teacher/quality/noise cases with 32 sampled actions each. It does not retrain students. Bound-violation counts by condition (none, bias 0.5, Gaussian 0.5, bias 1.5, Gaussian 1.5) are (9, 2, 0, 1, 0) in mechanics and (0, 3, 2, 6, 3) in reaction.

Task	Noise	Naive FP/pred.	Margin FP/pred.	Coverage (%)
Mechanical	none 0	2/480	2/480	100.0
Mechanical	bias 0.5	241/480	0/190	39.6
Mechanical	gaussian 0.5	296/480	0/98	20.4
Mechanical	bias 1.5	340/480	0/44	9.2
Mechanical	gaussian 1.5	413/480	0/24	5.0
Reaction	none 0	0/303	0/301	62.7
Reaction	bias 0.5	328/480	0/62	12.9
Reaction	gaussian 0.5	450/480	0/10	2.1
Reaction	bias 1.5	449/480	0/8	1.7
Reaction	gaussian 1.5	476/480	0/0	0.0

Table K.9: Independent local-action diagnostics over five teacher seeds and three check-point qualities: 480 action evaluations on 160 state/time points per task and noise condition, with states reused across teacher qualities. A fixed empirical curvature quantile comes from separate calibration trajectories. Ratios count false positive improvement predictions; zero predictions imply abstention. This calibrated margin is not a certified upper bound, and no student is retrained in this diagnostic.

Appendix L. Paired uncertainty

Task	Reference	Δ cost	95% CI	Wins
Mechanical	Cloning	-0.000179	[-0.016732, +0.016373]	2/5
Mechanical	Cached H	-0.038344	[-0.073294, -0.003394]	5/5
Mechanical	Selected fixed	+0.001000	[-0.007069, +0.009069]	2/5
Mechanical	Adaptive	-0.003630	[-0.023815, +0.016555]	3/5
Reaction	Cloning	-0.015024	[-0.071032, +0.040985]	3/5
Reaction	Cached H	-0.092694	[-0.161527, -0.023861]	5/5
Reaction	Selected fixed	-0.106005	[-0.255729, +0.043720]	5/5
Reaction	Adaptive	-0.001017	[-0.025796, +0.023762]	3/5

Table L.10: Paired arithmetic cost differences across five teacher/student/data replications. Intervals use a two-sided t calculation with four degrees of freedom; comparisons are exploratory and not multiplicity-adjusted.

Each audit interval concerns the five paired teacher/student/data replications. States within a trajectory and repeated actions under different

perturbations are not counted as independent replications. All selected policies and raw per-state costs are supplied, including runs worse than their teacher.

Appendix M. Reproducibility

The repository is https://github.com/yeoneung/teacher_student_hj. Its v1.0.0 release provides `research-artifacts-v1.zip`. Extracting this archive gives `research/README.md` for environment and execution instructions, `research/results/` for protocols, simulation outcomes and checkpoints, and the analysis scripts in `research/`. The release’s `artifact-manifest.json` records asset and file hashes. The reference archives `07_supporting_evidence.zip` and `08_data_and_checkpoints.zip`, available in the same release, are required by `verify_evidence.py --stage-inputs`.

The package supplies numerical-source and policy hashes, checkpoint eligibility, raw costs, diagnostic arrays, environment versions and entry points. Teacher construction, validation and test records remain separate. Analysis re-evaluates stored policies without retraining.

The loss intervention includes 600 final policies and separate initial and confirmation protocols. The confirmation plan hashes the initial report and both numerical entry points before training. Checks verify normal/radial decompositions and inactive-constraint parameter equality. The minimal-information study supplies 480 policies, 80 caches with actual query counts and a frozen protocol. Its analysis verifies source/cache/checkpoint hashes and recovered coefficients. Separate 120-case algebraic checks cover sub-bound indistinguishability and response error; these are not training replications.

The direct-query follow-up supplies 120 additional policies, all re-evaluated, and scalar-count/timing records for 85 caches. Four conditions reuse grid data for fidelity checks; the constructed shared-actuation condition uses new seeds. The 2,304 approximate-feedback checks are algebraic, not training replications. Across the distinct studies the paper reports 1,306 fitted policies; their samples and statistics are not pooled.

Appendix N. Set geometry and checks

Let $C = \{u : \|u\| \leq R_U, \|u - m\| \leq R\}$ be nonempty. For a source s , first compute its projections p_U and p_B onto the individual balls. If p_U lies in the second ball it is the projection onto C ; if p_B lies in the first ball it is

the projection. Otherwise both boundaries are active. With $d = \|m\| > 0$, $e = m/d$ and

$$h = \frac{R_U^2 + d^2 - R^2}{2d}, \quad s_\perp = s - (s^\top e)e, \quad (\text{N.1})$$

the intersection of the boundaries lies in $u^\top e = h$ and has radius $\sqrt{R_U^2 - h^2}$ about he . Its closest point is

$$\text{proj}_C s = he + \sqrt{R_U^2 - h^2} \frac{s_\perp}{\|s_\perp\|}. \quad (\text{N.2})$$

In the non-tangent both-active case $s_\perp$ is nonzero; otherwise a single-ball projection is already feasible. Tangency collapses the radius to zero. Concentric balls are handled by a single-ball case. Thus no iterative optimization or state-space grid is required for the tested geometry.

The implementation is compared with an independent SciPy SLSQP solution on 120 random feasible intersections in dimensions 2, 3, 8 and 32, with additional concentric, singleton and tangent cases. Central finite differences check the squared-distance gradient. These checks test the supplied geometry, not the validity of an empirical nonlinear curvature bound.

For a finite-horizon exact-model check, use $n = 8$, horizon 20, $x_{t+1} = Ax_t + Bu_t$, $A = 0.86I + 0.025(S + S^\top)$ with cyclic shift S , $B = 0.15I$, $Q = 0.1I$, $r = 0.07$, terminal matrix $P_{20} = I$, and $\|u\| \leq 2$. The globally feasible zero teacher has $P_t = Q + A^\top P_{t+1}A$ and value $x^\top P_t x$. Use its exact quadratic gradient and $\kappa_t = 2\lambda_{\max}(B^\top P_{t+1}B)$, the exact upper-model minimizer as anchor, $\alpha = 1/2$, and 128 standard-normal initial states from the documented random seed. Project the constant source 0.11 at every student-visited state.

The mean teacher cost is 3.329050 and the projected policy cost is 2.782772, a difference of -0.546278 , bounded above by -0.541921 in Eq. (E.8). Every evaluated action belongs to its set. The unprojected constant policy instead has cost 3.756661; its positive cost difference 0.427611 is below the corresponding distance bound 3.726794. The maximum telescoping error across both policies is below 3×10^{-15} . These are full-trajectory numerical checks of an analytic quadratic model, not learned-policy performance or a nonlinear certificate. The one-step constant-policy example in Appendix E independently checks strict approximation relaxation, its adverse cost comparison, and the singleton limit for an exact minimizer anchor.

Appendix O. Exploratory set comparisons

These one-pair-per-task comparisons report every relaxation, including unsuccessful runs. They are exploratory and do not supply confidence intervals across independent training runs. Each entry is the mean over 512 test initials. Table O.11 uses zero-residual analytic-base initialization; Table O.12 uses the pretrained width-64 student. The validation-only criterion and disjoint replication are specified in Appendix I.

Table O.11: Analytic-base initialization: test cost by task and student width. All mechanical students and all set students select their initial policy.

Method	Mechanics		Reaction	
	Width 16	Width 64	Width 16	Width 64
Point anchor	43.812646	43.812646	0.972505	1.325053
Adaptive point	43.812646	43.812646	0.972505	0.801105
Upper model	43.812646	43.812646	1.325053	1.325053
Set $\alpha = 0.25$	43.812646	43.812646	1.325053	1.325053
Set $\alpha = 0.50$	43.812646	43.812646	1.325053	1.325053
Set $\alpha = 0.75$	43.812646	43.812646	1.325053	1.325053

Table O.12: Pretrained-student pilot. Validation costs select relaxation and determine replication eligibility; test costs are not used for those decisions.

Method	Mechanics		Reaction	
	Validation	Test	Validation	Test
Point anchor	3.153080	3.148607	1.049305	0.854282
Adaptive point	3.130316	3.094025	1.049305	0.854282
Upper model	3.282828	3.236654	1.049305	0.854282
Set $\alpha = 0.25$	3.282828	3.236654	0.938238	0.993972
Set $\alpha = 0.50$	3.282828	3.236654	1.049305	0.854282
Set $\alpha = 0.75$	3.282828	3.236654	1.030278	0.950207

Training uses float32 neural updates and projection parameters; the independent diagnostic reconstruction and exact checks use float64. Across all 56 saved students, comparing float32 projections with the float64 formula on their diagnostic states gives maximum action error divided by the control-ball radius below 4.79×10^{-4} . This finite-precision check does not make the empirically bounded nonlinear model a certificate. Checkpoints, per-initial-state costs, sampled actions and source hashes accompany every run.

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